PURDUE POLYTECHNIC INSTITUTE
School of Engineering Technology

August 3, 2026

Dear Dr. Andrei Shkel,

We are pleased to submit our revision of the manuscript entitled "*Design of a Keel Bone Monitoring System in Egg Laying Hens*," authored by: *Nathan Huang, Omosalewa Marlene Fadayomi, Darrin M. Karcher, Walter Daniel Leon Salas and Robert A. Nawrocki*, for reconsideration as an Original Research Article in *IEEE Sensors Letters*. We have addressed all the concerns of all the reviewers.

Keel bone damage (KBD) is a prevalent and critical issue affecting a significant percentage of commercial laying hens worldwide, leading to decreased productivity and severe animal welfare concerns. Traditional assessment methods—such as manual analysis, radiography, and post-mortem examinations—fundamentally fail to capture the high-frequency, real-time kinetic events that cause initial damage. Our study bridges this gap by presenting the design, fabrication, to create a minimally obtrusive, low-power biomedical sensor node based on the Arduino Nicla Sense ME architecture.

This work introduces an ultra-miniature (22.94 mm × 22.94 mm) wireless sensor node capable of high-density integration of a 6-axis IMU and environmental sensors. It should be noted that these are commercially available sensors, and we have done the work of integrating all these environmental sensors in a larger system design, as well as developing the software to successfully collect data. This is significant as it provides a scalable framework for real-time biomechanical monitoring in animal science, where traditional devices are often too bulky, too massive or provide insufficient data integrity for long-term studies.

The authors have reviewed the comments of the reviewers and made the appropriate changes in the paper, as well as provided some commentary in the form of a rebuttal attached to the resubmission. Overall, the novelty of this system lies in its integrated hardware and firmware optimization, which utilizes custom power management strategies to achieve a reduction in average power consumption. Given that KBD is a time-sensitive welfare issue, the rapid development of this hardware architecture is critical. It provides the empirical impact data necessary for the redesign of cage systems, moving beyond the limitations of post-mortem results.

The authors have not previously published or submitted work on a similar topic in any archival journal. In comparison to existing methodologies, such as those by Casey-Trott et al., which rely on manual assessment, our system provides high-resolution, real-time kinetic profiling. Furthermore, while previous IoT frameworks for environmental monitoring often utilize larger platforms like the STM32 Nucleo, as in past research, our design achieves extreme miniaturization.

This paper has not been posted on any preprint repositories, including but not limited to IEEEXplore. Through analysis of the data collected by the device, we can monitor real-time events on animals to further the development of biomedical devices in animal science. We believe this contribution will be of strong interest to the broad readership of *IEEE Sensors Letters*.

We sincerely thank you for your time and reconsideration of our manuscript. We look forward to your response.

Sincerely:

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